

AEGIS - Distribution of Work

Module: COMP47250 Team Software Project (Microsoft partnership, 2026)
Project: AEGIS - Adaptive Exam Guardian and Integrity System
Repository: <https://github.com/comp47250-team-project/aegis>
Jira: <https://comp47250-ucd.atlassian.net> (project key `AEGIS`)
Team: Vijeth Prashanth Hegde, Raushan Nerkar, Tejas Tholakalabavi Dinesh, Lakshmi Kiran C M, Jeevika Bangalore Thanuj Kumar, Tejas Pantharapalya Venkatesh
Development window: 1 June - 7 August 2026

1. Purpose and agreement

This document records each team member's individual contributions to AEGIS, submitted alongside the final report to support grade individualisation. Every figure in this document is derived from the repository's own history and the live Jira board, and every figure can be reproduced with the commands in the following sections. This document evidences breadth and consistency of contribution per member: across code, infrastructure, evaluation, the report itself, and the presentation/demo, rather than asserting a ranked split.

2. How to verify (methodology)

All commands below run against `main` (`git log ... HEAD`) and reproduce every number in Sec. 3–5.

```
# Consolidated non-merge commit count per member (identities merged - see Sec. 3)
git log --no-merges --format="%ae" HEAD | grep -cE "vijeth\.ph@outlook\.com" # Vijeth
git log --no-merges --format="%ae" HEAD | grep -cE "raushannerkar" # Raushan
git log --no-merges --format="%ae" HEAD | grep -cE "tejas\.tholakalabavi@gmail\.com|tejas-td|Tejass-MacBook" # Tejas T D
git log --no-merges --format="%ae" HEAD | grep -cE "tejas6404@gmail\.com|Tejas1107" # Tejas P V
git log --no-merges --format="%ae" HEAD | grep -cE "jeevikabt27" # Jeevika
git log --no-merges --format="%ae" HEAD | grep -cE "lakshmikirancm" # Lakshmi

# Raw identities (why the regexes above are needed - two members are both named "Tejas",
# and several members committed under both a personal email and a GitHub noreply address)
git shortlog -sne --no-merges HEAD

# Distinct ISO weeks a member committed in, and their first/last commit date
# (checks the module's "commits must span the whole project, not cluster at the ends" criterion)
git log --no-merges --format="%ae|%ad" --date=format:"%Y-W%V" HEAD | grep -E "<pattern>" | cut -d'|' -f2 | sort -u
git log --no-merges --format="%ae|%ad" --date=short --reverse HEAD | grep -E "<pattern>"

# Pull requests merged into main: branch-merge commits ...
git log --merges --format="%ae|%s" HEAD | grep -v "Merge branch"
# ... plus squash-merge commits that reference "(#NN)" in the subject
git log --no-merges --format="%ae|%s" HEAD | grep "(#"
```

Reporting basis: "Commits" counts non-merge authored commits on `main` with split Git identities consolidated by email (Sec. 3). The Snyk bot and Dependabot commits (automated dependency bumps) are excluded from per-member figures. "PRs" counts both branch-merge commits and squash-merge commits.

3. Contribution summary

Member	Primary areas owned	Non-merge commits*	Jira tickets (of 126)	PRs merged (branch+squash)
Vijeth Prashanth Hegde <i>(team lead)</i>	Backend architecture, WebSocket gateway, telemetry persistence, exam/quiz APIs, deployment config, E2E tests	21	27	14
Raushan Nerkar	Azure infrastructure & IaC, CI/CD deployment, professor live-monitoring UI, super-admin console, manual grading & result release	23	26	23
Tejas Tholakalabavi Dinesh	Student exam shell & auth UI, countdown/auto-submit, event-timeline view, warning banners, quiz LRU cache	18	12	12
Lakshmi Kiran C M	JWT auth, RBAC, signal-scoring components, professor exam-create form, scoring presets, student groups, PR integration	16	16	16
Jeevika Bangalore Thanuj Kumar	Telemetry SDK core, keystroke/IKI signal detectors, pure signal-scorer engine, baseline calculator, TypeScript strict-mode	15	15	15
Tejas Pantharapalya Venkatesh	Database schema & ORM, Alembic migrations, seed data, risk-score triggers, CSV export, architecture decision record	14	12	12
Dependabot (automated)	Dependency version bumps	2	—	—
Total		109	126+	

* Non-merge authored commits on `main` with split Git identities consolidated (Sec. 2). + 126 Jira issues exist in total; 18 remain unassigned (mostly epics used as containers rather than work items, plus a handful of demo/report-logistics tasks) - see Sec. 4.

Sec. 4 evidences that all six members contributed across every category of project work, and no member's work is concentrated at the start or end of the trimester.

4. Coverage across all project components

Every member contributed to more than one category of project work, evidencing that AEGIS was not divided into six independent silos:

Member	Backend/API	Frontend	Infra/CI/CD	DB/schema	Evaluation	Report section	Presentation/demo	PR review & integration
Vijeth Prashanth Hegde	✓ (lead)	✓	✓			Abstract, System Design, LaTeX assembly	Slide 1,2,6,7,10,20	✓
Raushan Nerkar	✓	✓	✓ (lead)		✓ (load test)	Implementation	Slide 13,14,16	✓
Tejas Tholakalabavi Dinesh		✓ (lead)				Conclusion + report QA	Slide 3,4,18,19	✓ (7 PRs)
Lakshmi Kiran C M	✓	✓			✓ (usability study)	Related Work + Discussion	Slide 5,12,17	✓ (13 PRs)
Jeevika B. Thanuj Kumar	✓	✓ (SDK)			✓ (detection pilot)	Evaluation	Slide 11,15	
Tejas Pantharapalya Venkatesh	✓			✓ (lead)		Introduction + References	Slide 8,9	✓ (4 PRs)

5. Per-member narratives

Vijeth Prashanth Hegde (team lead)

Vijeth set up the repository and established the **backend architecture and real-time pipeline**: the professor/quiz/exam routes (AEGIS-33/34), the answer-submission endpoint and GDPR consent screen (AEGIS-35/38), the authenticated WebSocket gateway (AEGIS-48), and the Service Bus telemetry- persistence path (AEGIS-49): the design that keeps telemetry independent of exam submission. He drove deployment configuration (AEGIS-65), fixed a batch of multi-session bugs (AEGIS-110), added the Playwright E2E suite (AEGIS-109), and built a Kubernetes/AKS deployment path. As team lead he authored the CODEOWNERS map and `AGENTS.md`, coordinated sprint planning, and assembled the final report's LaTeX skeleton, writing the Abstract and System Design sections himself. His key architectural decision was enforcing that monitoring failures can never block a student's submission.

Raushan Nerkar

Raushan owned the **cloud and deployment backbone** of AEGIS. He authored the Azure Infrastructure-as- Code (AEGIS-21 resource group/budget/RBAC, AEGIS-63 full Bicep environment), the Key Vault + managed-identity secret wiring (AEGIS-77), and the ACR image deployment pipeline (AEGIS-66), including the CD role-assignment fixes that unblocked rollout. On the product side he built the professor live-monitoring dashboard (AEGIS-58/59), the super-admin console (AEGIS-107), the manual grading and conditional result-release workflow (AEGIS-112), question-bank import (AEGIS-90), and the health-check endpoint (AEGIS-89). He also drafted the report's Implementation section and led three demo scenes (stack proof, live monitor, admin console). A key decision he drove was consuming secrets exclusively from Key Vault via managed identity rather than baking them into images.

Tejas Tholakalabavi Dinesh

Tejas TD owned most of the **student-facing exam experience**: the login/register pages with JWT auth context (AEGIS-29), the student dashboard (AEGIS-37), the countdown timer with auto-submit (AEGIS-40), the post-submission confirmation page (AEGIS-41), and the non-blocking warning banners (AEGIS-85). He built the event-timeline history view (AEGIS-60), bulk CSV enrolment on the frontend (AEGIS-87), and the quiz LRU cache (AEGIS-108) after a mentor noted the exam page was "keeps hitting db" on refresh. He merged more pull requests than anyone besides Lakshmi (7 squash-merged branches plus reviewing teammates' work), drafted the report's Conclusion section, and led the whole-report QA pass. A notable technical lesson he owned was fixing a stale-closure bug in the auto-submit timer using ref-based state (AEGIS-40).

Lakshmi Kiran C M

Lakshmi contributed across **authentication, the signal-scoring layer, and PR integration**. She authored the JWT auth endpoints (AEGIS-26) and the RBAC dependency (AEGIS-27), then built several scoring components: the tab-blur/paste scorers (AEGIS-54), the first-keypress/answer-time/resize scorers (AEGIS-56), the scoring sensitivity presets (AEGIS-84), and student groups/cohorts (AEGIS-105): the feature a mentor requested directly ("no option to group students"). She merged **13 pull requests**, more than any other member, making her the team's primary integrator; she also ran the usability (SUS) study and drafted the report's Related Work and Discussion sections. Her footprint is heaviest in `backend/tests` and `backend/app/services`, reflecting a test-first approach to the scoring rules.

Jeevika Bangalore Thanuj Kumar

Jeevika owned the **telemetry SDK core and the pure scoring engine**: the most algorithmically involved, test-heavy modules. She scaffolded the SDK with the circular buffer and offline queue (AEGIS-42), authored the keystroke IKI signal detector (AEGIS-45), built the pure signal-scorer engine (AEGIS-52) and the baseline calculator (AEGIS-53) with dedicated unit tests for each, and implemented the IKI outlier scorer with z-score windowing (AEGIS-55). She also set up the pre-commit tooling (AEGIS-23), drove TypeScript strict-mode compliance (AEGIS-69), designed and ran the detection pilot that produced the report's headline evaluation numbers, and drafted the report's Evaluation section. Although her commit count is on the lower end, each commit delivered a self-contained, thoroughly tested module; her key decision was keeping the scorers as pure functions to make them deterministic and unit-testable.

Tejas Pantharapalya Venkatesh

Tejas PV built the **data foundation** everything else depends on: the initial SQLAlchemy ORM and database wiring (AEGIS-30), the seed script for users, professors, enrolments and quizzes (AEGIS-32), the risk-score and alert-flag triggers (AEGIS-57), and the CSV export API with integration tests (AEGIS-61). He authored the architecture decision record `DECISIONS.md` (AEGIS-91): the document that answers a mentor's direct question about technology trade-offs, and drafted the report's Introduction and References. His work centres on `backend/app/models`, `backend/app/database.py`, and the migration/seed tooling. A key contribution was establishing the schema and seed data early so the rest of the team could develop and test against realistic data.

6. Authorship of this document

Each member wrote and took ownership of their own paragraph in Sec. 5. The shared sections (1-4, 6) were compiled from the repository's commit history, the live Jira board, and the artefacts cited in Sec. 4, and were reviewed by all six members before submission. Every figure was individually verified against the repository as shown in Sec. 2, and every member confirms their own narrative in Sec. 5 accurately reflects their work.